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Models and Tools for Cyber-Physical Systems
Class sheet 2
WITH SOLUTIONS

Exercise 1: Modelling the Water Tank

Following the same procedure used in the lecture for the thermostat, draw a hybrid automaton for the water tank system depicted in Figure 1.

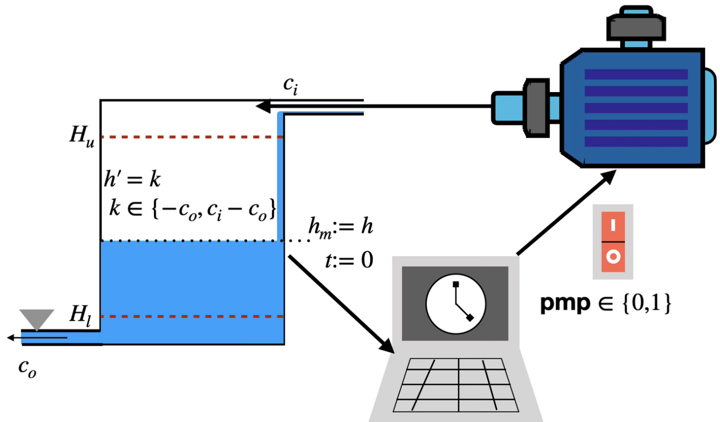

Figure 1: A water tank system to be modelled as a hybrid automaton.

. . . . Solution sketch . . . .

The completed hybrid automaton is shown in Figure 2.

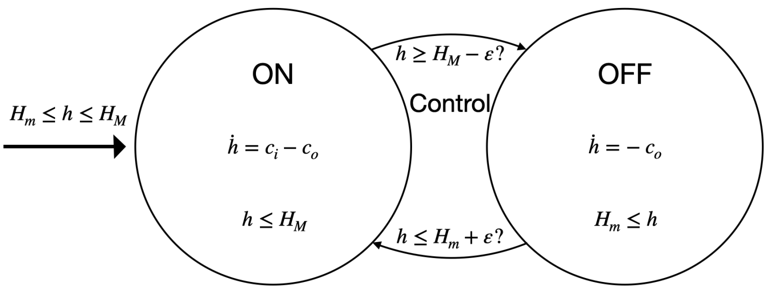

Figure 2: Hybrid automaton for the water tank.

Exercise 2: Modelling the Bouncing Ball

We model a ball dropped from rest as a hybrid system. The diagram in Figure 3 shows the hybrid automaton with four missing labels. Fill in the labels using the constraints below.

- The ball starts from rest, so the initial velocity is zero: $v(0) = v_0 = 0$.
- Let h_0 denote the starting height: $h(0) = h_0$.
- The rate of change of height equals the velocity, and the velocity decreases at the rate of the gravitational constant g: $\dot{h} = v, \quad \dot{v} = -g$.
- The ground is the boundary condition $h(t) \geq 0$.
- When the ball hits the ground ($h(t) = 0$), the velocity changes discretely via $v := -a \cdot v$, where $0 < a < 1$ is a damping constant.

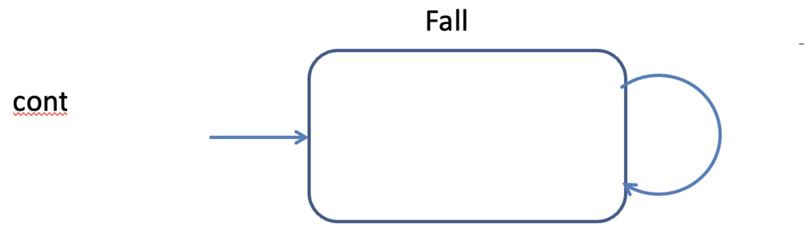

Figure 3: Hybrid automaton for the bouncing ball with four labels missing. Fill in: (1) the inputs, (2) the ODEs, (3) the invariant, (4) the transition reset.

...................................Solution sketch ....

The completed diagram is shown in Figure 4.

(a) **Inputs (initial values):** $h := h_0, \quad v := v_0$ (or $v := 0$).

(b) **ODEs:** $\dot{h} = v, \quad \dot{v} = -g$.

(c) **Invariant:** $h(t) \geq 0$.

(d) **Transition reset:** $h = 0 \rightarrow (bump!; v := -a \cdot v)$.

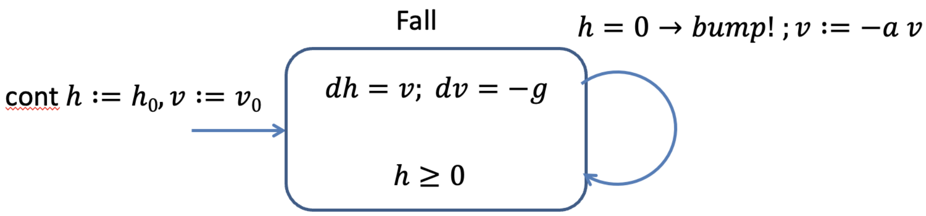

Figure 4: Completed hybrid automaton for the bouncing ball.

Exercise 3: Analysing the Bouncing Ball

Although the model from Exercise 2 abstracts the collision with the ground, it hides a subtle undesired behaviour. We analyse the system's evolution over time.
The general solution to the ODE $\dot{h} = v$, $\dot{v} = -g$ is:

$$v(t) = -g \cdot t + v_{prev}, \qquad h(t) = -\tfrac{g}{2} \cdot t^2 + v_{prev} \cdot t + h_{prev}.$$

(a) How long does it take the ball to reach the floor before the first bounce? Express your answer in terms of h_0 and g.

(b) Use the result from (a) to find the velocity v_1 immediately after the first bounce, in terms of h_0, a, and g.

(c) Find the time t_{next} between two successive bounces (from one bounce to the next). *Hint:* between bounces, $h_{prev} = h_{next} = 0$. Express your answer in terms of v_{prev} and g.

(d) From (c), deduce v_{next} in terms of v_{prev}.

(e) What is the velocity v_{n+1} immediately after the $(n+1)$th bounce, in terms of the velocity v_n after the nth bounce? And in terms of the velocity v_1 after the first bounce?

(f) Using your answers from (c) and (e), find the total duration of all bounces after the first one, $\displaystyle\sum_{i=1}^{\infty} t_{i+1}$. *Hint:* consider t_{n+1} from (c) as expressed in terms of v_{n+1} (why?) and recall $\displaystyle\sum_{i=0}^{\infty} r^i = \frac{1}{1-r}$ for $|r| < 1$.

(g) Does this make physical sense?

. . . . Solution sketch . . . .

(a) Before the first bounce, $v_{prev} = v_0 = 0$ and $h_{prev} = h_0$. Setting $h(t) = 0$:

$$0 = -\tfrac{g}{2} \cdot t^2 + h_0 \quad \Longrightarrow \quad t = \sqrt{\frac{2 \cdot h_0}{g}}.$$

(b) The velocity just before the first bounce is $v\left(\sqrt{2 \cdot h_0/g}\right) = -g \cdot \sqrt{2 \cdot h_0/g} = -\sqrt{2 \cdot h_0 \cdot g}$. After the bounce, $v := -a \cdot v$, so

$$v_1 = a \cdot \sqrt{2 \cdot h_0 \cdot g}.$$

(c) Between bounces $h_{prev} = h_{next} = 0$, so

$$0 = -\tfrac{g}{2} \cdot t^2 + v_{prev} \cdot t \;=\; t\big(v_{prev} - \tfrac{g}{2} \cdot t\big)\,.$$

The non-zero solution is

$$t_{next} = \frac{2 \cdot v_{prev}}{g}.$$

(d) Substituting t_{next} into the velocity equation:

$$v_{next} = -g \cdot \frac{2 \cdot v_{prev}}{g} + v_{prev} = -v_{prev}.$$

(e) After the bounce, $v := -a \cdot v_{next}$, so

$$v_{n+1} = -a \cdot (-v_n) = a \cdot v_n.$$

Hence $v_2 = a \cdot v_1$, $v_3 = a^2 \cdot v_1$, and in general $v_{n+1} = a^n \cdot v_1$.

(f) From (c) and (e):

$$\sum_{i=1}^{\infty} t_{i+1} = \sum_{i=1}^{\infty} \frac{2 \cdot v_{i+1}}{g} = \frac{2 \cdot v_1}{g} \sum_{i=1}^{\infty} a^i = \frac{2 \cdot v_1}{g} \cdot \frac{a}{1-a}.$$

Since $0 < a < 1$, this sum is *finite*.

(g) No, it is not physically possible for infinitely many bounces to occur in a finite amount of time.

Exercise 4: Zeno and Non-Zeno Definitions

Complete each sentence below.

- An infinite execution $s_0 \xrightarrow{t_1} s_1 \xrightarrow{t_2} \cdots$ is called *Zeno* if the infinite sum of all durations t_i is ____________________.
- An infinite execution $s_0 \xrightarrow{t_1} s_1 \xrightarrow{t_2} \cdots$ is called *non-Zeno* if the infinite sum of all durations t_i ____________________.
- A state s of a process HP is called *Zeno* if __________ (Hint: $\forall$) execution starting in s is Zeno.
- A state s of a process HP is called *non-Zeno* if ____________ (Hint: $\exists$) non-Zeno execution starting in s.
- A hybrid process HP is called *non-Zeno* if __________ (Hint: $\forall$) reachable state of HP is non-Zeno.
- In a non-Zeno system HP, it is possible for time to _______________ (Hint: $t \to \infty$) at every point during one of its executions.
- In a Zeno system HP, an execution could end up in a state from which only Zeno executions are possible, that is, an execution where the total duration of all transitions is _______________ (Hint: $\sum t < b$).

.... Solution sketch ....

- ... is **bounded by a constant** (alternatively, **finite**).
- ... **diverges** (i.e. $\sum t_i = \infty$).
- ... if **every** execution starting in s is Zeno.
- ... if **there exists** a non-Zeno execution starting in s.
- ... if **every** reachable state of HP is non-Zeno.
- ... time to **diverge** ($t \to \infty$)...
- ... total duration is **finite** (i.e. $\sum t_i < b$ for some constant b).

Exercise 5: Identifying Zeno Systems

For each system below, determine whether it is Zeno or non-Zeno. Justify your answer using the definitions from Exercise 4.

(a) The thermostat from the lecture.

(b) The bouncing ball from Exercise 2.

(c) The system in Figure 5.

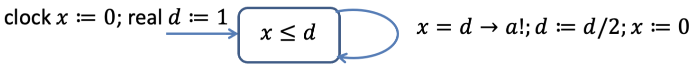

Figure 5: System (c).

(d) The system in Figure 6.

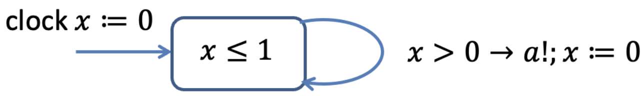

Figure 6: System (d).

(e) The system in Figure 7.

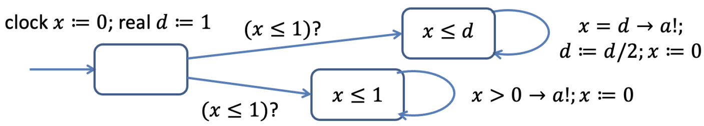

Figure 7: System (e).

(f) *Extra:* Does the system in (d) have a Zeno execution?

(g) *Extra:* What is the total duration of all transitions in system (c)?

.... Solution sketch ....

(a) **Non-Zeno.** The thermostat cycles between ON and OFF with each phase consistently adding a (bounded from below) positive duration. Thus, the total time diverges.

(b) **Zeno.** From Exercise 3(f) the sum of all bounce durations after the first is $\frac{2 \cdot v_1}{g} \cdot \frac{a}{1-a}$, which is finite. Infinitely many bounces occur in finite time.

(c) **Zeno.** Every execution of this system is Zeno: the total duration of all transitions is finite (see (g)).

(d) **Non-Zeno.** The guard $x \leq 1$ and the self-transition $x > 0 \rightarrow x := 0$ add a strictly positive amount of time less than 1. Every reachable state s could fire up a sequence of events of duration $1/2$ ($s \stackrel{1/2}{\rightarrow} s_1 \stackrel{1/2}{\rightarrow} \dots$) So the HP is non-Zeno.

(e) **Zeno.** There are reachable states, from which all trajectories have a Zeno behaviour, namely, those states that non-deterministically end-up in the upper branch that mimics Figure 5

(f) Yes. Although the system in (d) is non-Zeno, every reachable state admits a trajectory where the durations reduce geometrically, ($s \stackrel{1/2}{\rightarrow} s_1 \stackrel{1/4}{\rightarrow} s_2 \stackrel{1/8}{\rightarrow} \dots$) possibly mimicking the pattern of (c) and producing a Zeno trajectory.

(g) System (c) starts at $d = 1$ and resets $d := d/2$ each time the clock reaches d. The total duration is

$$1 + \tfrac{1}{2} + \tfrac{1}{4} + \cdots = \sum_{n=0}^{\infty} \left(\tfrac{1}{2}\right)^n = 2.$$

Exercise 6: Why Zeno Processes Matter

Recall the following definitions: a state s is *reachable* if there exists a finite execution starting in an initial state and ending in s. A property φ is an *invariant* if all reachable states satisfy φ.

(a) Is mode B in Figure 8 reachable?

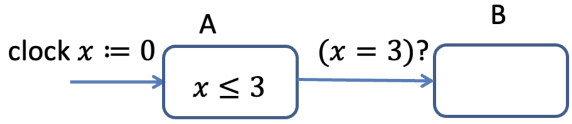

Figure 8: System for question (a).

(b) Is mode B in Figure 9 reachable?

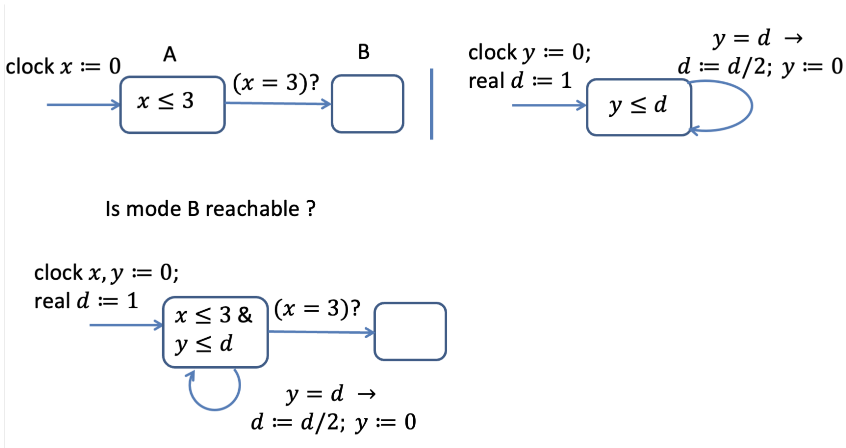

Figure 9: System for question (b): a composition involving a Zeno process.

(c) To prevent the bouncing ball from being Zeno, add a discrete command that fires when the velocity has "diminished enough": instead of bouncing, the ball comes to rest by setting $v := 0$. Extend the diagram from Exercise 2 with this modification, using e for the "small-enough" threshold.

................................Solution sketch ....

(a) **Yes.** At 3 units of time, Mode B is reachable.

(b) **No.** The composition includes a Zeno process from before that leads to infinitely many actions before 2 units of time pass. Once time is "stuck", the clock variable x cannot attain the value 3 and transition to B. This shows that the presence of a Zeno process in a composed system can make states of other processes unreachable.

(c) The extended non-Zeno bouncing ball is shown in Figure 10.

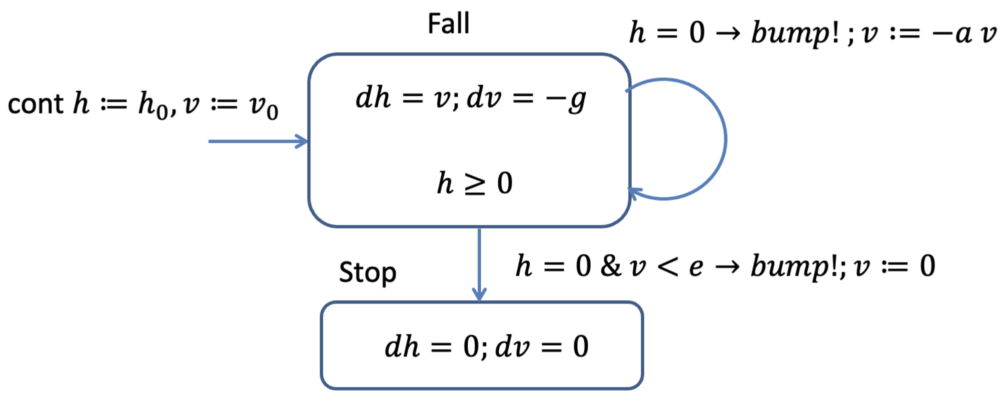

Figure 10: Non-Zeno bouncing ball: a rest mode prevents infinite bounces.